Project Milestone & Full Development Report

Authoritative Summary of Architecture, Portals, RVM Integration & Completed Features

Repository: github.com/hameedshaik2345/OpenBin_WebApplication	**Stack:** React 19 + Node/Express + PostgreSQL + Firebase
Primary Branch: main (Synchronized)	**Status:** All Features Functional & Mobile-Optimized

1. Executive Summary & Solution Mission

OpenBin is an enterprise-grade circular economy platform bridging physical Reverse Vending Machines (RVMs), campus consumers, FMCG brands (EPR compliance), and authorized waste recyclers. The platform incentivizes sustainable behavior by dispensing real-time financial rewards (OpenCoins) when users recycle plastic (PET) and aluminium beverage containers, while providing brands and regulatory bodies (CPCB) with verifiable, audit-ready EPR recovery records.

2. Detailed Work Completed to Date

A. Consumer & Public User Portal (/dashboard)

- **Unified Mobile Navigation:** Engineered an intuitive 5-tab mobile navigation bar with Home, Wallet, floating centered Scan button, History, and Impact. Removed legacy duplicate navbars and relocated Profile to an icon-only button in the top header.
- **Dual-Mode QR Claim Engine:** Embedded a live browser camera scanner using Html5Qrcode with auto-environmental lens detection, paired with a resilient JSON payload manual paste fallback for testing.
- **Anti-Replay Security & Cryptographic Settlement:** Claim tokens are hashed via SHA-256 upon RVM issuance. Double-claiming, expired tokens, or malformed payloads trigger graceful contextual error handling.
- **Double-Entry Wallet & Ledger:** Users can view live INR balances, initiate real-time UPI withdrawals, or donate to the Clean Campus Community Fund. Historical credit/debit movements are dynamically rendered from PostgreSQL.
- **Schema Normalization:** Resolved critical database column mismatches (claim_status, estimated_weight_g, reward_value) ensuring real deposits accurately increment personal impact counters and history records.
- **Dynamic Impact Metrics & Charts:** Replaced static demo figures with true PostgreSQL counts for items recycled, landfill weight diversion (kg), and dynamic monthly Chart.js visualization.

B. Brand Partner & Enterprise EPR Portal (/epr)

- **Role-Based Access Control:** Route guards securely isolate Brand Partner functionality to EPR and ADMIN roles.
- **Real-Time Socket.io Telemetry:** Subscribed to liveState WebSocket broadcasts, streaming physical machine fullness, bottle counts, and operational health in real time.
- **SKU & Material Analysis:** Track recovery volumes classified by manufacturer (e.g., PepsiCo, Coca-Cola), brand (Aquafina, Kinley), and material type (PET, Aluminium).
- **Compliance Certification & Exports:** Generates certified quarterly CPCB-compliant recovery certificates with automated CSV and PDF download capabilities.
- **Full Mobile Responsiveness:** Comprehensive touch navigation and responsive tables so brand managers can monitor recycling quotas from mobile devices.

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C. Administrative Operations Suite (/admin)

- **AppShell Responsive Drawer:** Added a slide-out hamburger navigation drawer on mobile and persistent sidebar on desktop, featuring an unread notification center.
- **User & Partner Management (/admin/users):** Complete administrative search, user role modification, account status freezing/unfreezing, and dedicated Brand Partner onboarding form.
- **Smart RVM Fleet (/admin/rvms):** Provision physical or simulated machines, assign GPS coordinates/campus buildings, and monitor storage fullness.
- **Scrap Material Pricing (/admin/pricing):** Dynamic reimbursement rate controller for recyclables per kilogram with immutable historical rate freezing.
- **Collection Logistics (/admin/collection):** Log collection batches picked up from campus RVMs, register accredited recycler facilities, and generate reconciliation audit logs.
- **Immutable Audit Trail (/admin/audit):** System-wide immutable logging of all financial, pricing, and administrative events.

D. Hardware Integration & RVM Simulator (/rvm-simulator)

- **Complete IoT State Machine:** Replicates physical RVM sensor cycles (IDLE → CONTAINER_INSERTED → CLASSIFYING → SESSION_READY → DISPENSING).
- **Dynamic QR Token Generation:** Live generation of high-contrast QR codes embedding UUID transaction tokens with configurable 15-minute expiration timers.
- **Automated Expiry Worker:** Background cron worker running in Node.js that checks for expired unclaimed rewards every 60 seconds and transfers them to community charity.
- **Restyled Emerald & Navy Theme:** Modernized bezel with high contrast, glowing status indicators, and clean Manrope typography matching OpenBin branding.

3. Technical Architecture & Endpoints Matrix

Module / Domain	Routes & Components	Key Responsibilities & Data Handlers
Auth & Identity	/login, /register /api/auth/*	Firebase token verification, PostgreSQL user record synchronization, role dispatching (USER, EPR, ADMIN).
User Portal	/dashboard, /dashboard/scan /dashboard/history	Real-time wallet, QR scanner modal, withdrawal & donation execution, transaction filtering, dynamic charts.
Brand ERP (EPR)	/epr, /epr/rvms /epr/reports	Live machine state streaming via Socket.io, brand SKU tracking, quarterly CPCB report exports.
Admin Operations	/admin/* (9 Modules)	User management, fleet provisioning, dynamic pricing, batch reconciliation, and system audit logs.
Hardware / IoT	/rvm-simulator /api/rvm/transactions	Sensor telemetry ingestion, dynamic QR code generation, 15-minute TTL enforcement, background expiry worker.

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4. Verified Active Credentials & Role Access

All administrative and enterprise accounts have been verified directly against active Firebase Auth and PostgreSQL ledgers:

Account Role	Email Address	Password	Designated Landing Portal
Primary Administrator	openbinsolutions@gmail.com	openbin123	/admin (Full Authority)
Brand Partner (EPR)	partner@openbin.com	openbin123	/epr (Brand Dashboard)
Brand Partner (Alt)	partner@coolfizz.com	openbin123	/epr (Brand Dashboard)
Consumer / Public	Open self-registration	User-defined	/dashboard (User Portal)

5. Deployment Roadmap & Infrastructure Recommendations

Why Not Vercel for Backend: OpenBin's backend requires persistent TCP WebSockets (Socket.io) for streaming RVM machine states and continuous background intervals (expiryWorker.js) to expire unclaimed rewards. Serverless platforms like Vercel freeze idle instances and terminate WebSocket connections.

Recommended Cloud Architecture:

- **Frontend:** Deploy to **Vercel** (automatic builds from frontend root, fast CDN, zero config).
- **Backend API & WebSockets:** Deploy to **Render.com** or **Railway.app** as a standard Web Service (native 24/7 Node process, full Socket.io support, background worker persistence).
- **Relational Database:** Deploy to **Neon.tech** or **Supabase** (managed PostgreSQL, automated SSL connection pooling).
- **Authentication:** Whitelist the production Vercel domain in Firebase Console under Authentication → Authorized Domains.

6. Project Audit & Build Verification Status

✓ **Frontend Build:** npm run build succeeded with 0 errors (Vite production bundle generated).

✓ **Mobile Responsiveness:** Validated on 375px, 414px, and 768px viewport widths with responsive tables, drawer nav, and geometric bottom bar alignment.

✓ **Git Synchronization:** All commits, merge conflict resolutions, and schema bug fixes pushed cleanly to GitHub repository main branch.

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